

Takuji Usui

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ACADEMIC POSITIONS

Assistant Professor | 2027 –
Department of Ecology & Evolutionary Biology
Yale University

EEB Postdoctoral Research Fellow | Oct. 2025 – Current
University of Toronto
Collaborators: Prof. Stephen Wright & Prof. Megan Frederickson

Faculty of Arts & Sciences Postdoctoral Fellow | 2023 – 2025
University of Toronto
Supervisor: Prof. Megan Frederickson

EDUCATION

University of British Columbia | 2017 – 2023
PhD, Botany
Thesis: Experimental evolution of species ranges and coexistence
Supervisor: Prof. Amy Angert

University of Edinburgh | 2011 – 2015
BSc, Evolutionary Biology
Thesis: Temporal shifts and temperature sensitivity of spring migratory phenology
Supervisor: Prof. Ally Phillimore

RESEARCH & FIELD TRAINING

Research Studentship | 2016 – 2017
University of New South Wales & University of Sydney
Supervisors: Prof. Shinichi Nakagawa & Prof. Alistair Senior

Research Studentship | 2015 – 2016
University of Edinburgh
Supervisor: Prof. Ally Phillimore

Field Assistant | 2015 – 2016

University of Edinburgh

Supervisor: Prof. Jarrod Hadfield

Tropical Biology Association Field Research | 2015

Kibale National Park, Uganda

Tropical Biology Association

PUBLICATIONS

† = Co-first authors; * Co-senior authors

Pre-prints:

17. Urquhart, C.A., **Usui, T.**, Angert, A.L. & Williams, J.L. (2026) Eco-evolutionary dynamics are shaped by competition in experimental range expansions. Preprint available at: doi.org/10.64898/2026.06.01.729372.
16. **Usui, T.†**, Sakarchi, J.†, Duchon, P., Hart, S.P., Turcotte, M.M., Xu, S., Angert, A.L.* & Germain, R.M.* (2026) Niche differentiation confers coexistence prior to the species boundary in an aquatic plant. Preprint available at: doi.org/10.64898/2026.04.03.716288.
15. Marjakangas, E-L., Barreto, E., Borregaard, M.K., Chaikin, S., Carvajal-Quintero, J.D., Marshall, K.E., **Usui, T.**, Pires, M.M., Graham, C.H. & Lessard, J-P. (2026) The role of biotic permeability in species' range dynamics.

Published:

14. **Usui, T.†**, Yu, J.† & Frederickson, M.F. (2026) For colonization success, should hosts and microbes travel alone, together, or swap partners along the way? Accepted at *New Phytologist*.
13. **Usui, T.** & Angert, A.L. (2026) Competition enables rapid adaptation to a warming range edge in a model plant community. *Science*, 392, 6802.
12. Gillies, G.J., Angert, A.L. & **Usui, T.** (2024) Temperature-dependence and genetic variation in resource acquisition strategies in a model freshwater plant. *Functional Ecology*, 38, 1600-1610.
11. **Usui, T.** & Angert, A.L. (2024) Range expansion is both slower and more variable with rapid evolution across a spatial gradient in temperature. *Ecology Letters*, 27, e14406.
10. **Usui, T.†**, Lerner, D.†, Angert, A.L., Eckert, I., Garroway, C.J., Hargreaves, A.L., Lancaster, L.T., Lessard, J-P., Riva, F., Schmidt, C., van der Burg, K. & Marshall, K. (2023) The evolution of plasticity at geographic range edges. *Trends in Ecology & Evolution*, 38, 831-842.
9. **Usui, T.**, MacLeod, M., McCann, S., Senior, A.* & Nakagawa, S.* (2021) Meta-analysis of variation suggests that embracing heterogeneity improves both replicability and generalizability in preclinical research. *PLoS Biology*, 19(5), e3001009.

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8. Bontrager, M., **Usui, T.**, Lee-Yaw, J.A., Anstett, D.N., Branch, H.A., Hargreaves, A.L. & Angert, A.L. (2021) Adaptation across geographic ranges is consistent with strong selection in marginal climates and legacies of range expansion. *Evolution*, 75, 1316-1333.
7. Germain, R.M., Angert, A.L., Bassar, R.D., Henao Diaz, F.L., Hart, S.P., Otto, S.P., Rolland, J., Sakarchi, J., Schluter, D., Siepielski, A.M., Turcotte, M.M., **Usui, T.** & Waters, M. (2020) On the origin of coexisting species. *Trends in Ecology & Evolution*, 36, 284-283.
6. Miller, T.E.X., Angert, A.L.A., Brown, C.D., Lee-Yaw, J.A., Lewis, M., Lutscher, F., Marculis, N.G., Melbourne, B.A., Shaw, A.K., Szűcs, M., Tabares, O., **Usui, T.**, Weiss-Lehman, C. & Williams, J.L. (2020) Eco-evolutionary dynamics of range expansion. *Ecology*, 101, e03139.
5. Baur, J., Laurentino, T.G., **Usui, T.** & Eichhorn M. (2019) Liana abundance and relationships to sapling and tree hosts in an East African primary forest. *African Journal of Ecology*, 57, 130-136.
4. **Usui, T.** & Angert, A.L.A. (2018) Species Ranges. *Elsevier Reference Module in Life Sciences*.
3. **Usui, T.**, Noble, D.W.A., O'Dea, R.E., Fangmeier, M.L., Lagisz, M., Hesselson, D. & Nakagawa, S. (2018) The French Press: a repeatable and high-throughput approach to exercising zebrafish (*Danio rerio*). *PeerJ*, e4292.
2. Fangmeier, M., Noble, D.W.A., O'Dea, R.E., **Usui, T.**, Lagisz, M., Hesselson, D. & Nakagawa, S. (2018) Computer animation technology in behavioral sciences: a sequential, automatic, and high-throughput approach to quantifying personality in zebrafish (*Danio rerio*). *Zebrafish*, 15, 206-210.
1. **Usui, T.**, Butchart, S.H.M. & Phillimore, A.B. (2017) Temporal shifts and temperature sensitivity of avian spring migration phenology: a phylogenetic meta-analysis. *Journal of Animal Ecology*, 86, 250-261. Select press coverage: **BBC News; The Guardian**

FELLOWSHIPS & GRANTS

2025	EEB Postdoctoral Research Fellowship University of Toronto	\$156,000
2023	Faculty of Arts & Science Postdoctoral Fellowship University of Toronto	\$96,000
2017	International Doctoral Fellowship University of British Columbia	\$138,000
2017	Scientia PhD Scholarship University of New South Wales (<i>Declined</i>)	\$200,000
2017	Graduate Research Scholarship University College London (<i>Declined</i>)	£52,660

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2017	Overseas Research Scholarship University College London (<i>Declined</i>)	£78,480
2017	NERC Science and Solutions for a Changing Planet Doctoral Training Partnership Imperial College London (<i>Declined</i>)	£62,935
2016	Research Studentship University of New South Wales	\$14,000
2015	Research Studentship University of Edinburgh	£2500

COMPETITIVE AWARDS & HONOURS

2024	Evolution Early Career Award University of Wisconsin-Madison	\$150
2023	Excellence in Doctoral Research Award Canadian Society for Ecology & Evolution	\$600
2021	Best Talk Award Botany & Zoology Conference, University of British Columbia	-
2021	Student Research Award American Society of Naturalists	\$2000
2019	GREG Lewontin Award Society for the Study of Evolution	\$2000
2018	Vladimir J Krajina Prize in Plant Ecology University of British Columbia	\$2000
2018	Best Poster Award Brackendale Ecology & Evolution Conference	-
2015	Ashworth Medal and Prize for highest grade in Evolutionary Biology Honours University of Edinburgh	£250

INVITED PRESENTATIONS

(^ΨAward Talks; ‡ Symposium Talks)

2026	ASN Awards Symposium, Evolution Annual Online Meeting ^Ψ
2026	<i>EEB Departmental Seminar</i> , Yale University
2025	<i>EEOB Departmental Seminar</i> , University of California-Riverside
2025	<i>Biology Colloquium</i> , University of Wisconsin-Madison
2025	<i>Biology Department Seminar</i> , Boston University
2025	<i>Integrative Biology Departmental Seminar Series</i> , University of Guelph

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- 2024 *Evolution Early Career Award*, University of Wisconsin-Madison^Ψ
2024 *Linking theory to empirical research*, CSEE Annual Conference[‡]
2024 *Duckweeds, microbes, & symbiosis*, University of British Columbia[‡]
2023 *Doctoral Awards Symposium*, CSEE Annual Conference^Ψ
2022 *A new frontier in symbiosis research*, University of Toronto[‡]
2021 *Lessons for the lab from meta-research*, Centre for Open Science[‡]

CONTRIBUTED PRESENTATIONS

([†] Posters; [‡] Talks; ^Ψ Awards)

- 2026 CSEE Annual Conference, ON
2026 Atwood Colloquium, University of Toronto, ON[‡]
2023 EEB Departmental Seminar Series, University of Toronto, ON[‡]
2023 Botany Departmental Seminar Series, University of British Columbia, BC[‡]
2023 Asilomar Conference, Society of American Naturalists, CA[‡]
2022 Joint Annual Meeting of ESA & CSEE, QB[‡]
2021 Botany & Zoology Annual Meeting, University of British Columbia, BC^{‡Ψ}
2019 Brackendale Ecology & Evolution Conference, BC[‡]
2018 Brackendale Ecology & Evolution Conference, BC^{†Ψ}
2017 Brackendale Ecology & Evolution Conference, BC[†]

SYNERGISTIC RESEARCH ACTIVITIES

Invited member: **Thermal plasticity and range dynamics**, German Centre for Integrative Biodiversity Research (iDiv) | 2026

Organizer: International Range Edge Dynamics (RED) Working Group

Invited member: **Climate variability at range edges**, German Centre for Integrative Biodiversity Research (iDiv) | 2025

Organizer: International Range Edge Dynamics (RED) Working Group

Invited member: **Temporal variation in dispersal and range expansion**, CIEE Working Group at University of Toronto | 2025

Organizers: Graydon Gillies (Memorial U), Mia Akbar (U Toronto), Carissa Brown (Memorial U)

Lead member of working group: **Plasticity across species' ranges**, UBC | 2023

Organizers: Katie Marshall (UBC) and Jean-Philippe Lessard (Concordia University)

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Invited member of working group: **Coexistence on evolutionary scales**, UBC | 2019
Organizers: Rachel Germain (UBC) and Adam Siepielski (University of Arkansas)

Invited member of working group: **Eco-evo dynamics of range expansion**, UBC | 2018
Organizers: Tom Miller (Rice University) and Jennifer Williams (UBC)

TEACHING & MENTORING

Teaching Assistant | 2020 – 2023

BIOL 121: Genetics, Ecology & Evolution

BIOL 300: Biostatistics

BIOL 336: Fundamentals of Evolution (**Lead/Senior TA** 2021 and 2023)

BIOL 406: Plant Ecology (**Lead TA in course design** 2021; **Lead TA** 2022-2023)

Undergraduate Supervision

Jake Kilgour (Centre for Global Change Science Summer Internship Award, 2026)

Gemma Hornsey (Centre for Global Change Science Summer Internship Award, 2025)

Gabrielle Diez (Work-learn, 2024-2025)

Jerry Yu (Work-learn 2023; KSR USRA 2024; Honours student, 2024-2025)

Joshua Dyer (NSERC Undergraduate Student Research Award, 2023)

Amy Wong (Work-learn, 2022 & 2023)

Emma Pflanz (Research Mentorship Program, 2022)

Burak Ozkan (Research volunteer, 2022)

Kenzie Chute (Research volunteer, 2022)

Sourena Oveisi (Work-learn, 2022)

Vicky Kim (Research volunteer, 2022)

Graydon Gillies (NSERC Undergraduate Student Research Award, 2020 & 2021)

Chris Spangenberg (Work-learn, 2020 & 2021)

Hafsa Ahmed, UBC (Work-learn, 2019)

Emma Menchions, UBC (Work-learn, 2018)

SERVICE ACTIVITIES

Manuscript reviews

American Journal of Botany; American Naturalist; Aquatic Botany; Ecological Monographs; Ecology; Ecography; Evolution; Global Change Biology; Journal of Animal Ecology; Journal of Avian Biology; Journal of Biogeography; Journal of Evolutionary Biology; Nature Ecology & Evolution; Oikos; Perspectives in Plant Ecology; PLOS Computational Biology; Evolution and Systematics

Professional memberships

American Naturalist Society; Botanical Society of America; Canadian Society for Ecology & Evolution; Ecological Society of America; National Biodiversity Network; Society for the Study of Evolution; Tropical Biology Association

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Academic service

- 2026 Conference Planning Committee
Canadian Society for Ecology & Evolution
- 2026 Judge for EEB Undergraduate Research Poster Fair
University of Toronto
- 2025 Member of EDI Committee
Canadian Society for Ecology & Evolution
- 2025 Postdoctoral Representative
Department of Ecology & Evolutionary Biology, University of Toronto
- 2025 Postdoctoral Councillor
Canadian Society for Ecology & Evolution
- 2024 Judge Panel for Excellence in Doctoral Awards
Canadian Society for Ecology & Evolution
- 2021 Conference Planning Committee
Canadian Society for Ecology & Evolution
- 2020 Graduate Student Representative for Biodiversity PDF Selection Committee
University of British Columbia
- 2020 Selection Committee for Annual Evolution Lecture
University of British Columbia
- 2019 Lead Organizer and Host for Annual Evolution Lecture
University of British Columbia

SELECT MEDIA FEATURES

- 2024 **Computing and the future of AI in EEB (by Julia Boyle, University of Toronto)**
The EEQ Quarterly, University of Toronto
- 2023 **Using experimental evolution of duckweeds to understand spatiotemporal patterns of biodiversity**
International Duckweed Forum
- 2022 **Teaching Spotlight: Getting to know diverse scientists (by Christine Goedhart, UBC)**
BioNews, University of British Columbia
- 2020 **How species distributions are moving in response to climate change**
Researchers Revealed, University of British Columbia
- 2016 **Bird migration and global warming**
World Service News, BBC
- 2016 **Early birds miss out**
The Times

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- 2016 **Climate change driving birds to migrate early, research reveals**
The Guardian
- 2016 **Global temperature rises causing birds to migrate early**
The Independent